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DC Motor Slow-Speed Troubleshooting Worksheet

Measurement procedure and speed budget for PMDC and geared motors running below their expected speed

Take voltage, current and speed readings at the same moment, at the motor terminals, and record them against the motor serial number. The speed budget in the table shows how to turn those readings into a clear split of where the lost rpm went. Isolate and lock out the supply before uncoupling the motor or touching brush gear.

Step 1 — Collect reference data

- [ ] Nameplate voltage, current, rated speed and power; duty type.
- [ ] No-load speed and current from the test sheet or commissioning record.
- [ ] Gear ratio on record and the gearbox plate ratio actually fitted.
- [ ] Controller type and settings: maximum speed, IR compensation, current limit, reference range (0–5 V or 0–10 V).

Step 2 — No-load measurements

- [ ] Run the machine empty or decouple the motor; record terminal voltage (meter in DC mode), current and speed.
- [ ] On geared motors measure output speed and multiply by the ratio; a mismatch with motor speed reveals a wrong ratio.
- [ ] Slow at no load points to supply voltage, speed reference, winding variant or gear ratio.
- [ ] Faster than the original no-load speed points to weakened magnets; test the back-EMF constant cold.

Step 3 — Loaded measurements

- [ ] Record output speed, motor terminal voltage, controller output voltage and current at the same moment.
- [ ] Controller output minus terminal voltage = cable and connection drop; aim for about 3 % of nominal or less at rated current.
- [ ] Current above nameplate means overload or drag: turn the driveline by hand with the motor isolated.
- [ ] Repeat at the start and end of a shift on battery equipment, and from a cold start on cold mornings.

Step 4 — Controller and reference checks

- [ ] Measure the speed reference at the controller terminals with the operator control at full scale.
- [ ] Confirm 0–5 V or 0–10 V scaling matches the controller input setting.
- [ ] Check the maximum-speed trimmer and output voltage at full reference against the motor's rated voltage.
- [ ] Confirm IR compensation is enabled and set for this motor; over-compensation causes hunting.

Step 5 — Build the speed budget

- [ ] Ke = (V − I0 × R − brush drop) ÷ ω0 from no-load test data, with ω0 = 2π × N0 ÷ 60.
- [ ] Speed lost to voltage shortfall = 9.549 × (rated V − measured V) ÷ Ke rpm.
- [ ] Speed lost to extra current = 9.549 × (measured I − rated I) × R ÷ Ke rpm.
- [ ] Predicted speed = rated speed − both losses; compare with measured speed.
- [ ] A residual of more than a few per cent indicates extra internal resistance (brushes, commutator) or slip in belts and couplings.

Step 6 — Mechanical and brush gear checks

- [ ] Brush length against the wear limit, spring force, commutator film colour and evenness.
- [ ] Locked-rotor test at reduced voltage: apparent resistance should match the original at the same current.
- [ ] Belt tension and slip: driven speed × pulley ratio should match motor speed within a few per cent.
- [ ] Gearbox lubricant grade suited to the minimum ambient temperature; brake fully released.

Speed budget (worked example, 180 V, 3.5 A, 1,750 rpm PMDC motor)

Line item	Basis	Value	Speed effect
Back-EMF constant Ke	(180 − 0.35 × 5.6 − 2) ÷ 204.2 rad/s	0.862 V·s/rad	—
Predicted speed at rated 3.5 A	(180 − 19.6 − 2) ÷ 0.862	183.8 rad/s	1,755 rpm
Voltage shortfall	180 V rated − 168 V measured	12 V	−133 rpm
Extra load current	(4.2 − 3.5) A × 5.6 Ω	3.9 V	−43 rpm
Predicted speed in service	1,755 − 133 − 43	—	1,578 rpm
Measured speed	Tachometer	—	1,520 rpm
Unexplained residual	58 rpm × 0.862 ÷ 9.549	5.2 V, about 1.25 Ω at 4.2 A	−58 rpm: check brushes

9.549 = 60 ÷ 2π converts rad/s to rpm. Figures are an illustration; use your motor's own nameplate, test data and measured resistance.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.